Лабораторная работа

Номер 16

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Информация

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Цель работы

Целью данной работы является получение навыков настройки VPN-туннеля через незащищённое Интернет-соединение.

Размещаем в рабочей области проекта в соответствии с модельными предположениями оборудование для сети Университета г. Пиза. (Рис. 1.1).

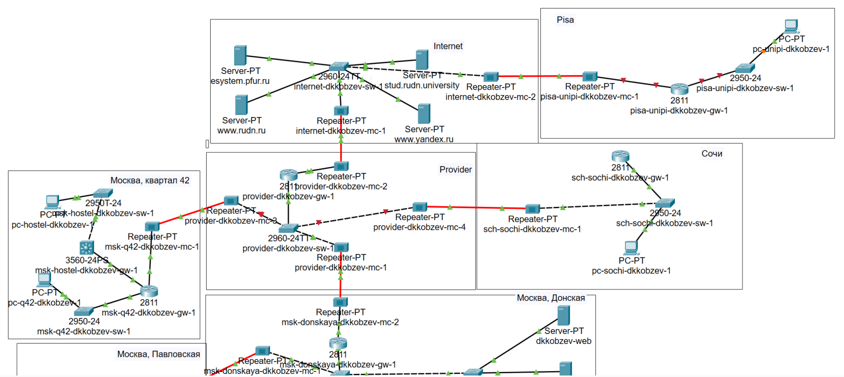

Рис. 1: Схема сети с дополнительными площадками

Настройка VPN

В физической рабочей области проекта создаем город Пиза, здание Университета г. Пиза. Перемещаем туда соответствующее оборудование. (Рис. 1.2).

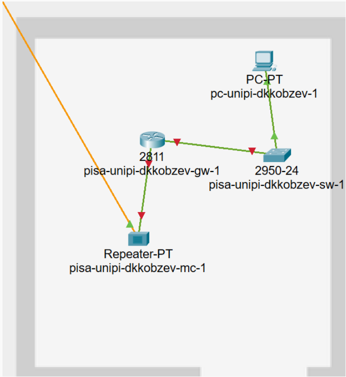

Рис. 2: Физическая рабочая область

Настройка площадки в г. Пиза

Выполняем первоначальную настройку и настройку интерфейсов оборудования сети Университета г. Пиза (Рис. 1.3), (Рис. 1.4), (Рис. 1.5), (Рис. 1.6).

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname pisa-unipi-dkkobzev-gw-1
pisa-unipi-dkkobzev-gw-1(config)#line vty 0 4
pisa-unipi-dkkobzev-gw-1(config-line)#password cisco
pisa-unipi-dkkobzev-gw-1(config-line)#login
pisa-unipi-dkkobzev-gw-1(config-line)#exit
pisa-unipi-dkkobzev-gw-1(config)#line console 0
pisa-unipi-dkkobzev-gw-1(config-line)#password cisco
pisa-unipi-dkkobzev-gw-1(config-line)#login
pisa-unipi-dkkobzev-gw-1(config-line)#exit
pisa-unipi-dkkobzev-gw-1(config)#enable secret cisco
pisa-unipi-dkkobzev-gw-1(config)#service password-encryption
pisa-unipi-dkkobzev-gw-1(config)#username admin privilege 1 secret cisco
pisa-unipi-dkkobzev-gw-1(config)#ip domain-name unipi.edu
pisa-unipi-dkkobzev-gw-1(config)#crypto key generate rsa
The name for the keys will be: pisa-unipi-dkkobzev-gw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-dkkobzev-gw-1(config)#line vty 0 4
*Mar 1 0:18:28.338: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:18:28.338: %SSH-5-ENABLED: SSH 1.5 has been enabled
pisa-unipi-dkkobzev-gw-1(config-line)#transport input ssh
pisa-unipi-dkkobzev-gw-1(config-line)#exit
```

Рис. 3: Первоначальная настройка маршрутизатора pisa-unipi-gw-1

Настройка площадки в г. Пиза

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname pisa-unipi-dkkobzev-sw-1
pisa-unipi-dkkobzev-sw-1(config)#line vty 0 4
pisa-unipi-dkkobzev-sw-1(config-line)#password cisco
pisa-unipi-dkkobzev-sw-1(config-line)#login
pisa-unipi-dkkobzev-sw-1(config-line)#exit
pisa-unipi-dkkobzev-sw-1(config)#line console 0
pisa-unipi-dkkobzev-sw-1(config-line)#password cisco
pisa-unipi-dkkobzev-sw-1(config-line)#login
pisa-unipi-dkkobzev-sw-1(config-line)#exit
pisa-unipi-dkkobzev-sw-1(config)#enable secret cisco
pisa-unipi-dkkobzev-sw-1(config)#service password-encryption
pisa-unipi-dkkobzev-sw-1(config)#username admin privilege 1 secret cisco
pisa-unipi-dkkobzev-sw-1(config)#ip domain-name unipi.edu
pisa-unipi-dkkobzev-sw-1(config)#crypto key generate rsa
The name for the keys will be: pisa-unipi-dkkobzev-sw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-dkkobzev-sw-1(config)#line vty 0 4
*Mar 1 0:20:52.977: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:20:52.977: %SSH-5-ENABLED: SSH 1.5 has been enabled
pisa-unipi-dkkobzev-sw-1(config-line)#transport input ssh
pisa-unipi-dkkobzev-sw-1(config-line)#exit
```

Рис. 4: Первоначальная настройка коммутатора pisa-unipi-sw-1

Настройка площадки в г. Пиза

```
pisa-unipi-dkkobzev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-dkkobzev-gw-1(config)#int f0/0
pisa-unipi-dkkobzev-gw-1(config-if)#no shutdown

pisa-unipi-dkkobzev-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

pisa-unipi-dkkobzev-gw-1(config-if)#exit
pisa-unipi-dkkobzev-gw-1(config)#int f0/0.401
pisa-unipi-dkkobzev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

pisa-unipi-dkkobzev-gw-1(config-subif)#encapsulation dot1Q 401
pisa-unipi-dkkobzev-gw-1(config-subif)#ip address 10.131.0.1 255.255.255.0
pisa-unipi-dkkobzev-gw-1(config-subif)#description unipi-main
pisa-unipi-dkkobzev-gw-1(config-subif)#exit
pisa-unipi-dkkobzev-gw-1(config)#int f0/1
pisa-unipi-dkkobzev-gw-1(config-if)#no shutdown

pisa-unipi-dkkobzev-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

pisa-unipi-dkkobzev-gw-1(config-if)#ip address 192.0.2.20 255.255.255.0
pisa-unipi-dkkobzev-gw-1(config-if)#description internet
pisa-unipi-dkkobzev-gw-1(config-if)#exit
pisa-unipi-dkkobzev-gw-1(config)#ip route 0.0.0.0 0.0.0.0 192.0.2.1
pisa-unipi-dkkobzev-gw-1(config)#exit
```

Рис. 5: Настройка интерфейсов маршрутизатора pisa-unipi-gw-1

Настройка площадки в г. Пиза

```
pisa-unipi-dkkobzev-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-dkkobzev-sw-1(config)#int f0/24
pisa-unipi-dkkobzev-sw-1(config-if)#switchport mode trunk

pisa-unipi-dkkobzev-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

pisa-unipi-dkkobzev-sw-1(config-if)#exit
pisa-unipi-dkkobzev-sw-1(config)#int f0/1
pisa-unipi-dkkobzev-sw-1(config-if)#switchport mode access
pisa-unipi-dkkobzev-sw-1(config-if)#switchport access vlan 401
% Access VLAN does not exist. Creating vlan 401
pisa-unipi-dkkobzev-sw-1(config-if)#exit
pisa-unipi-dkkobzev-sw-1(config)#vlan 401
pisa-unipi-dkkobzev-sw-1(config-vlan)#name unipi-main
pisa-unipi-dkkobzev-sw-1(config-vlan)#exit
pisa-unipi-dkkobzev-sw-1(config)#int vlan401
pisa-unipi-dkkobzev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

pisa-unipi-dkkobzev-sw-1(config-if)#no shutdown
```

Рис. 6: Настройка интерфейсов коммутатора pisa-unipi-sw-1

Настраиваем VPN на основе протокола GRE (Рис. 1.7), (Рис. 1.8).

```
msk-donskaya-dkkobzev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-dkkobzev-gw-1(config)#int Tunnel0

msk-donskaya-dkkobzev-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

msk-donskaya-dkkobzev-gw-1(config-if)#ip address 10.128.255.253 255.255.255.252
msk-donskaya-dkkobzev-gw-1(config-if)#tunnel source f0/1.4
msk-donskaya-dkkobzev-gw-1(config-if)#tunnel destination 192.0.2.20
msk-donskaya-dkkobzev-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

msk-donskaya-dkkobzev-gw-1(config-if)#exit
msk-donskaya-dkkobzev-gw-1(config)#int loopback0

msk-donskaya-dkkobzev-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

msk-donskaya-dkkobzev-gw-1(config-if)#ip address 10.128.254.1 255.255.255.255
msk-donskaya-dkkobzev-gw-1(config-if)#exit
msk-donskaya-dkkobzev-gw-1(config)#ip route 10.128.254.5 255.255.255.255 10.128.255.254
```

Рис. 7: Настройка маршрутизатора msk-donskaya-gw-1

Настройка VPN на основе GRE

```
pisa-unipi-dkkobzev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-dkkobzev-gw-1(config)#int Tunnel0

pisa-unipi-dkkobzev-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

pisa-unipi-dkkobzev-gw-1(config-if)#ip address 10.128.255.254 255.255.255.252
pisa-unipi-dkkobzev-gw-1(config-if)#tunnel source f0/1
pisa-unipi-dkkobzev-gw-1(config-if)#tunnel destination 198.51.100.2
pisa-unipi-dkkobzev-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

pisa-unipi-dkkobzev-gw-1(config-if)#exit
pisa-unipi-dkkobzev-gw-1(config)#int loopback0

pisa-unipi-dkkobzev-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

pisa-unipi-dkkobzev-gw-1(config-if)#ip address 10.128.254.5 255.255.255.255
pisa-unipi-dkkobzev-gw-1(config-if)#exit
pisa-unipi-dkkobzev-gw-1(config)#ip route 10.128.254.1 255.255.255.255 10.128.255.253
pisa-unipi-dkkobzev-gw-1(config)#router ospf 1
pisa-unipi-dkkobzev-gw-1(config-router)#router-id 10.128.254.5
pisa-unipi-dkkobzev-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
pisa-unipi-dkkobzev-gw-1(config-router)#exit
```

Рис. 8: Настройка маршрутизатора pisa-unipi-gw-1

Выводы

В результате выполнения лабораторной работы мною были получены навыки настройки VPN-туннеля через незащищённое Интернет-соединение.